Introduction

Every scientific experiment conducted to investigate the patterns in natural phenomenon may result with "events" which may or may not happen. Most of the events in real life have uncertainty in their happening. For example,

Disintegration of a given atom of radium in a given time interval may or may not disintegrate;

A plant may or may not be infected by species during rainy season; The event of increase in the gold price under an economic condition in a country; A drug administered to a cancer patient for curing a disease in a period of time.

In all these cases, there is an amount of uncertainty prevails. Even for a student, asking a particular question in the examination from a particular portion of a subject is uncertain. Yet, the student is compelled to take a decision during the preparation of examination, whether to go for an in-depth study or leaving the question in choice. In a nutshell, one has to take a wise decision under the conditions of uncertainty. In such a situation, knowledge about the chance or probability for occurrence of an event of interest is vital and calculation of probability for happening of an event is imperative.

It is very much essential to determine a quantitative value to the chance or probability for the occurrence of random events in many real life situations. Before defining probability, students gain knowledge on the following essential terms.

8.1 Random experiment, Sample space, Sample point

Experiment: In Statistics, by the word experiment it means 'an attempt to produce a result'. It need not be a laboratory experiment.

Random Experiment: If an experiment is such that

(i) all the possible outcomes of the experiment are predictable, in advance

(ii) outcome of any trial of the experiment is not known, in advance, and

(iii) it can be repeated any number of times under identical conditions, is called a random experiment.

Sample space: The set of all possible outcomes of a random experiment is called the sample space of the experiment and is usually denoted by S (or Ω). If S contains only finite number of elements, it is termed as finite sample space. If S contains countable number of elements, S may be called as countable sample space or discrete sample space. Otherwise, S is called an uncountable sample space.